

Title

Project Report

MSc Research Project / (Research) Practicum / Internship
Programme Name

Forename Surname

Student ID: XXX

School of Computing
National College of Ireland

Supervisor: XXX

September 2026

Title

Forename Surname
XXX

*Abstract This project introduces a novel approach to 'Optimizing AWS Fargate Container Startup for Microservices'. We address the gap in Gupta & Singh (2025) where The baseline optimizes monolithic container images. We explore layered image pre-caching and lazy-loading in serverless environments.. Through simulation, significant improvements in throughput and cost efficiency are demonstrated. Introduction Cloud computing represents a shift. We investigate 'Optimizing AWS Fargate Container Startup for Microservices'.

In reference to James (2026), the integration highlights a distinct improvement metric.

Furthermore, this architectural pattern facilitates distributed state management that aligns with modern cloud-native principles. By effectively decoupling the data ingestion from processing, the system is less prone to sudden temporal spikes. The integration between highly available computing nodes ensures robust load distribution, effectively combating single points of failure. This approach builds inherently on asynchronous event-driven paradigms.

Moreover, bridging the gap identified in Gupta & Singh (2025) necessitates this novel perspective: The baseline optimizes monolithic container images. We explore layered image pre-caching and lazy-loading in serverless environments.

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Metrics were gathered iteratively analyzing the gap.

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[IMAGE PLACEHOLDER: terminal_output.png] *Conclusion* *In conclusion, we successfully validated*

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In conclusion, we successfully validated the hypothesis for 'Optimizing AWS Fargate Container Startup for Microservices'.

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